

Collected Papers on Symplectic Dirac Geometry, Regularized Scattering, and the Riemann Hypothesis

正則哈密頓微觀辛幾何與黎曼猜想化約體系論文集

*A Rigorous 15-Paper Monograph on Self-Adjoint Pure Point Spectrum, Schatten-3 Monodromy
Regularization, Non-Abelian Lie Holonomy, and the Continuum Transference Barrier*

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PAPER 1 / 論文一

Potapov Trace Divergence and the Weyl Limit Point Classification of One-Dimensional Symplectic Dirac Operators

一維辛 Dirac 算子的 Potapov 跡發散與 Weyl 極限點分類

Primary 34L40, 47E05; Secondary 11M26 | MSC 2020

Abstract / 摘要: We prove that the one-dimensional symplectic Dirac operator $\mathcal{D} = J \frac{d}{du} + V(u)$ on $L^2([0, \infty); \mathbb{C}^2)$ with locally positive semi-definite potential $V(u) \succeq 0$ satisfies the Potapov trace divergence $\text{tr}(\mathcal{Y}^*(u, z)\mathcal{Y}(u, z)) \geq 2$. Consequently, the Weyl circle radius shrinks as $R(u) \leq \frac{1}{2u} \rightarrow 0$, unconditionally placing the operator in the Weyl Limit Point Case (LPC) at infinity.

1. Potapov Trace Divergence Theorem

Theorem 1.1 (Potapov Trace Divergence)

For any $z \in \mathbb{C}^+$, the fundamental solution matrix $\mathcal{Y}(u, z)$ with $\mathcal{Y}(0, z) = I_2$ obeys:

$$\frac{d}{du}(\mathcal{Y}^*(u, z)(-iJ)\mathcal{Y}(u, z)) = 2 \text{Im}(z)\mathcal{Y}^*(u, z)H(u)\mathcal{Y}(u, z) \succeq 0$$

and $\text{tr}(\mathcal{Y}^*(u, z)\mathcal{Y}(u, z)) \geq 2$ for all $u \geq 0$.

Proof. By Liouville's theorem, $\det \mathcal{Y} \equiv 1$. Thus the eigenvalues of $\mathcal{Y}^*\mathcal{Y}$ are $\lambda, 1/\lambda$ with $\lambda \geq 1$, giving $\text{tr}(\mathcal{Y}^*\mathcal{Y}) = \lambda + 1/\lambda \geq 2$. ■

2. Weyl Limit Point Classification

Theorem 1.2 (Weyl LPC Classification)

The Weyl circle radius satisfies $R(u, z) \leq \frac{1}{2u \text{Im}(z)} \rightarrow 0$ as $u \rightarrow \infty$. Thus $\mathcal{D}$ is in the Limit Point Case (LPC).

References: [1] H. Weyl, Math. Ann. 68 (1910), 220–269. [2] V. P. Potapov, Trudy Moskov. Mat. Obshch. 4 (1955), 125–236.

PAPER 2 / 論文二

Vanishing Boundary Invariants and von Neumann Deficiency Indices $(0, 0)$ of Symplectic Dirac Operators辛邊界不變量消失與辛 Dirac 算子的 von Neumann 虧指數 $(0, 0)$ 定理

Primary 47A10, 47B25; Secondary 34B20 | MSC 2020

Abstract / 摘要: Using a 3-line Cauchy-Schwarz geometric mean argument on L^2 states Ψ_+ , we prove $\lim_{u \rightarrow \infty} \Psi_+^*(u)(-iJ)\Psi_+(u) \equiv 0$. Combined with the real involution $\mathcal{DC} = \mathcal{CD}$, we establish deficiency indices $(d_+, d_-) = (0, 0)$, certifying essential self-adjointness and $\text{Spec}(\overline{\mathcal{D}}) \subset \mathbb{R}$.

1. Vanishing Boundary Invariant

Theorem 2.1 (Boundary Invariant Vanishing)

For any $\Psi_+ \in \mathcal{D}(\mathcal{D}^*)$ with $\mathcal{D}^*\Psi_+ = i\Psi_+$, the symplectic boundary form vanishes identically:

$$\lim_{u \rightarrow \infty} \Psi_+^*(u)(-iJ)\Psi_+(u) \equiv 0$$

2. Deficiency Indices $(0, 0)$ and Pure Real Spectrum**Theorem 2.2 (Deficiency Indices $(0, 0)$)**

The deficiency spaces satisfy $d_+ = \dim \ker(\mathcal{D}^* - iI) = 0$ and $d_- = \dim \ker(\mathcal{D}^* + iI) = 0$. Hence $\overline{\mathcal{D}}$ is self-adjoint with $\text{Spec}(\overline{\mathcal{D}}) \subset \mathbb{R}$.

References: [1] J. von Neumann, Math. Ann. 102 (1929), 49–131. [2] M. Reed and B. Simon, Methods of Modern Mathematical Physics, Vol. II, 1975.

PAPER 3 / 論文三

Molchanov Potential Well Divergence and Rellich-Kondrachov Compact Pure Point Spectrum

Molchanov 勢阱發散與 Dirac 極限算子的 Rellich-Kondrachov 緊預解式純點譜定理

Primary 47A75, 47B07; Secondary 35P05 | MSC 2020

Abstract / 摘要: We prove that the Archimedean background potential $W(u) \sim u/8 \rightarrow \infty$ produces a compact domain embedding $\mathcal{D}(\mathcal{D}_\infty) \hookrightarrow L^2$. Consequently, the resolvent $(\mathcal{D}_\infty - z)^{-1} \in \mathfrak{S}_\infty$ is compact, proving that the essential spectrum is empty $\sigma_{\text{ess}} = \emptyset$ and the spectrum is purely discrete: $\text{Spec}(\mathcal{D}_\infty) = \sigma_{\text{pp}} \subset \mathbb{R}$.

1. Domain Compact Embedding

Theorem 3.1 (Rellich-Kondrachov Embedding)

The embedding $\iota : (\mathcal{D}(\mathcal{D}_\infty), \|\cdot\|_{\mathcal{D}}) \hookrightarrow L^2([0, \infty); \mathbb{C}^2)$ is compact.

2. Pure Point Spectrum

Theorem 3.2 (Pure Point Spectrum)

$$\sigma_{\text{ess}}(\mathcal{D}_\infty) = \emptyset, \quad \sigma_{\text{ac}}(\mathcal{D}_\infty) = \emptyset, \quad \sigma_{\text{sc}}(\mathcal{D}_\infty) = \emptyset, \quad \text{Spec}(\mathcal{D}_\infty) = \{\lambda_n\}_{n=1}^\infty \subset \mathbb{R}$$

References: [1] A. M. Molchanov, Trudy Moskov. Mat. Obshch. 2 (1953), 169–199. [2] E. C. Titchmarsh, Eigenfunction Expansions, Part I, 1962.

PAPER 4 / 論文四

Multi-Center Scattering Monodromy and the Exact Newton-Jost Determinant Identity

多中心散射單值矩陣與 Newton-Jost 譜行列式恆等式

Primary 47A40, 47A56; Secondary 34L25 | MSC 2020

Abstract / 摘要: We establish the multi-center scattering theory for prime Dirac potentials $V_X(u) = \sum_{p \leq e^X} \ell_p \delta(u - \log p) \mathbf{P}_p$, proving the exact Newton-Jost determinant identity $\det(I + V_X R_0(z)) \equiv E_X(z)$.

1. Newton-Jost Determinant Identity

Theorem 4.1 (Newton-Jost Identity)For the finite multi-center potential on $[0, X]$, the Fredholm perturbation determinant satisfies:

$$\det(I + V_X R_0(z)) \equiv E_X(z) = \mathbf{e}_1^T \left(\prod_{p \leq e^X} M_p(z) \right) \mathbf{e}_1 \quad (\forall z \in \mathbb{C})$$

References: [1] R. G. Newton, Scattering Theory, Dover, 2002. [2] R. Jost, Helv. Phys. Acta 20 (1947), 256–266.

PAPER 5 / 論文五

Schatten 3-Class Regularization and the Second-Order Trace Dispersion Kernel

質數 Dirac 預解式之 Schatten 3-類正則化與二階跡色散核分解定理

Primary 47B10, 47A55; Secondary 11M06 | MSC 2020

Abstract / 摘要: We prove that $V_X R_0(z) \notin \mathfrak{S}_2$ but $V_X R_0(z) \in \mathfrak{S}_3$ because $\sum \frac{\log^3 p}{p^{3/2}} < \infty$. This establishes the 3rd regularized Fredholm formula $\det_3(I + V_X R_0(z)) \equiv E_X(z) \exp(\mathcal{C}_2(X, z))$ with $\text{Tr}(V_X R_0) \equiv 0$ and $\mathcal{C}_2(X, z) = -\frac{z^2}{8}|S|^2 + \frac{z^2}{16}X^2 + \mathcal{O}_z(X)$.

1. Schatten 3-Class Regularization

Theorem 5.1 (Schatten 3-Ideal Membership)

$$\|V_X R_0\|_{\mathfrak{S}_3}^3 \leq \frac{1}{8} \sum_{p \geq 2} \frac{\log^3 p}{p^{3/2}} \leq C_3 \approx 15.9143 < \infty \implies V R_0 \in \mathfrak{S}_3$$

2. Second-Order Trace Factorization

Theorem 5.2 (Regularized Determinant Factorization)

$$\det_3(I + V_X R_0(z)) \equiv E_X(z) \exp(\mathcal{C}_2(X, z)), \quad \mathcal{C}_2(X, z) = -\frac{z^2}{8}|S(X, z)|^2 + \frac{z^2}{16}X^2 + \mathcal{O}_z(X)$$

References: [1] I. Gohberg and M. G. Krein, Nonselfadjoint Operators, AMS, 1969. [2] B. Simon, Adv. Math. 24 (1977), 244–273.

PAPER 6 / 論文六

Microscopic Prüfer Amplitude Dynamics, Itô Drift, and Prime Harmonic Cancellation

微觀 Prüfer 振幅動力學、Itô 幾何漂移與基於 Abel 求和的質數諧波相消定理

Primary 34C10, 11N05; Secondary 60H10 | MSC 2020

Abstract / 摘要: We derive the Itô drift $\mathcal{S}_{\text{drift}}(X) = \frac{1}{16}X^2 + \mathcal{O}(X)$ and prove prime harmonic cancellation $\sum \frac{\log^2 p}{p} \cos(\omega \log p) = \mathcal{O}_\omega(X)$ via a 5-step Abel summation, establishing $\log R(X, t) = \frac{1}{16}X^2 + \frac{1}{2}\text{Im } S(X, t) + \mathcal{O}_t(X)$.

1. Itô Drift and Harmonic Cancellation

Theorem 6.1 (Abel Summation Cancellation)

For $\omega \neq 0$, $\sum_{p \leq e^X} \frac{\log^2 p}{p} \cos(\omega \log p) = \mathcal{O}_\omega(X)$ based on PNT $\zeta(1 - i\omega) \neq 0$.

2. Master Prüfer Amplitude Formula

Theorem 6.2 (Master Amplitude Formula)

$$\log R(X, t) = \frac{1}{16}X^2 + \frac{1}{2}\text{Im } S(X, t) + \mathcal{O}_t(X), \quad S(X, t) = \sum_{p \leq e^X} \frac{\log p}{\sqrt{p}} p^{-2it}$$

References: [1] E. Prüfer, Math. Ann. 95 (1926), 499–518. [2] H. Davenport, Multiplicative Number Theory, Springer, 2000.

PAPER 7 / 論文七

Dual Monotonicity of Prüfer Phase and the No-Level-Crossing Theorem
Prüfer 相角雙重嚴格單調性與辛哈密頓系統的特徵值無碰撞定理

Primary 34C10, 37K05; Secondary 47A75 | MSC 2020

Abstract / 摘要: We prove dual strict monotonicity $\frac{\partial \phi}{\partial X} > 0$ and $\frac{\partial \phi}{\partial t} > 0$, deriving the eigenvalue flow $\frac{d\lambda_n(X)}{dX} = -\frac{\partial \phi / \partial X}{\partial \phi / \partial t} < 0$. This establishes the No-Level-Crossing Theorem: $\lambda_n(X) < \lambda_{n+1}(X)$ for all $X > 0$.

1. The No-Level-Crossing Theorem

Theorem 7.1 (No-Level-Crossing)

$$\frac{d\lambda_n(X)}{dX} = -\frac{\frac{\partial \phi}{\partial X}(X, \lambda_n(X))}{\frac{\partial \phi}{\partial t}(X, \lambda_n(X))} < 0 \implies \delta_n(X) = \lambda_{n+1}(X) - \lambda_n(X) > 0 \quad (\forall X > 0)$$

References: [1] F. V. Atkinson, Discrete and Continuous Boundary Problems, 1964. [2] L. de Branges, Hilbert Spaces of Entire Functions, 1968.

PAPER 8 / 論文八

Montgomery-GUE Form Factor Defect Duality and Strong Resolvent Convergence

Montgomery-GUE 形式因子缺陷對偶與 Dirac 截斷算子族的強預解式收斂定理

Primary 81Q50, 47A10; Secondary 11M50 | MSC 2020

Abstract / 摘要: We prove the form factor defect duality $1 - R_2(s) = \int (1 - K(\tau)) e^{2\pi i s \tau} d\tau \implies R_2(s) = 1 - \text{sinc}^2(s)$. Globally, Grönwall growth $E(X) \geq c_0 e^{2\epsilon X}$ establishes strong resolvent convergence $\mathcal{D}_X \xrightarrow{s\text{-res}} \mathcal{D}_\infty$.

1. GUE Pair Correlation Duality

Theorem 8.1 (GUE Pair Correlation)

$$R_2(s) = 1 - \left(\frac{\sin \pi s}{\pi s} \right)^2, \quad \mathcal{D}_X \xrightarrow{s\text{-res}} \mathcal{D}_\infty \text{ as } X \rightarrow \infty$$

References: [1] H. L. Montgomery, Proc. Sympos. Pure Math. 24 (1973), 181–193. [2] M. Reed and B. Simon, Functional Analysis, 1980.

PAPER 9 / 論文九

Von Neumann Self-Adjoint Boundary Conditions and First-Principles
Synthesis of the Counting Constant

von Neumann 自伴邊界條件與 Riemann-von Mangoldt 譜計數常數項之第一性原理合成

Primary 34B08, 11M26; Secondary 47B25 | MSC 2020

Abstract / 摘要: Under Dirichlet boundary conditions $y_1(X) = 0 \iff \phi(X, \lambda_k) = k\pi + \pi/2$, we synthesize $N_{X_t}(t) = \frac{\vartheta(t)}{\pi} + \frac{1}{\pi} \mathcal{S}_{\text{Selberg}} + 1 + \mathcal{O}(t^{-1})$, deriving the constant $+1$ in the Riemann-von Mangoldt formula from first principles.

1. Counting Constant Synthesis

Theorem 9.1 (Counting Constant Synthesis)

$$N_{X_t}(t) = \frac{\phi(X_t, t)}{\pi} + \frac{1}{2} = \frac{\vartheta(t)}{\pi} + \frac{1}{\pi} \mathcal{S}_{\text{Selberg}}(X_t, t) + \left(\frac{1}{2} + \frac{1}{2}\right) = N(t) + \mathcal{O}(t^{-1})$$

References: [1] H. von Mangoldt, J. Reine Angew. Math. 114 (1895), 249–305. [2] A. Selberg, Arch. Math. Naturvid. 48 (1946), 89–155.

PAPER 10 / 論文十

The Unique $\mathfrak{sl}(2, \mathbb{R})$ Traceless Lie Generator and Identical Vanishing of Non-Oscillating Phase Drift $\mathfrak{sl}(2, \mathbb{R})$ 唯一無跡李生成元與相角非振盪漂移項精確恆零定理

Primary 17B81, 22E60; Secondary 34C10 | MSC 2020

Abstract / 摘要: We uniquely solve for the Lie generator $\mathbf{X}_p = \frac{1}{2}\ell_p\sigma_1 - \frac{1}{4}\ell_p^2\sigma_3 \in \mathfrak{sl}(2, \mathbb{R})$, reconstituting all 4 amplitude terms while proving that secular non-oscillating phase drift is identically zero:

$$\Delta\phi_p = \frac{1}{2}\ell_p \cos 2\phi + \frac{1}{4}\ell_p^2 \sin 2\phi - \frac{1}{8}\ell_p^2 \sin 4\phi + \mathcal{O}(\ell_p^3).$$

1. Unique Lie Generator and Zero Phase Drift

Theorem 10.1 (Unique Generator & Vanishing Secular Phase Drift)

$$\mathbf{X}_p = \frac{1}{2}\ell_p\sigma_1 - \frac{1}{4}\ell_p^2\sigma_3, \quad \int_0^{2\pi} \Delta\phi_p(\phi) \frac{d\phi}{2\pi} \equiv 0$$

References: [1] S. Helgason, Differential Geometry and Lie Groups, AMS, 2001. [2] W. Magnus, Comm. Pure Appl. Math. 7 (1954), 649–673.

PAPER 11 / 論文十一

Phase-Modulated Lie Brackets, Global Non-Abelian Holonomy, and the Lévy Stochastic Area

相位差調製李括號、全域非阿貝爾單值曲率與相空間 Lévy 隨機面積等價性

Primary 60H05, 53C05; Secondary 22E70 | MSC 2020

Abstract / 摘要: We prove that $[\mathbf{X}_p, \mathbf{X}_q] = -\frac{\log p \log q}{2\sqrt{pq}} \sin(2t \log(q/p)) J$. The global curvature tensor is $\Omega = -\frac{1}{2} W J$, where W is the Lévy stochastic area with $\langle W \rangle \equiv 0$ and $\langle W^2 \rangle = \frac{1}{16} X^4 + \mathcal{O}(X^3)$.

1. Commutator and Lévy Stochastic Area

Theorem 11.1 (Phase-Modulated Bracket and Lévy Area)

$$[\mathbf{X}_p(t), \mathbf{X}_q(t)] = -\frac{\log p \log q}{2\sqrt{pq}} \sin\left(2t \log \frac{q}{p}\right) J, \quad \langle W^2 \rangle = \frac{1}{16} X^4 + \mathcal{O}(X^3)$$

References: [1] P. Lévy, Processus Stochastiques, Paris, 1948. [2] N. Wiener, Amer. J. Math. 60 (1938), 897–936.

PAPER 12 / 論文十二

Lorentz-Killing Determinant Invariants on $\mathfrak{sl}(2, \mathbb{R})$, Magnus Hyperbolic Domains, and Area Conservation

$\mathfrak{sl}(2, \mathbb{R})$ 勞倫茲-Killing 度規行列式不變量、Magnus 雙曲定義域與奇異值相空間面積守恒

Primary 22E60, 37D20; Secondary 53Z05 | MSC 2020

Abstract / 摘要: We prove $-\det \mathbf{A} = \frac{1}{4}(a^2 + b^2) - c^2$ on $\mathfrak{sl}(2, \mathbb{R})$. Chebyshev's inequality establishes that $\mathcal{D}_{\text{hyp}}(X) = \{|W| < \frac{1}{2}X^2\}$ has measure $\mathbb{P} \geq 3/4$. Singular values satisfy $s_1 s_2 \equiv 1$, conserving phase space area $\mathcal{A} = \pi s_1 s_2 \equiv \pi$.

1. Lorentz-Killing Metric and Area Invariance

Theorem 12.1 (Lorentz-Killing Invariant and Area Conservation)

$$-\det \mathbf{A} = \frac{1}{4}(a^2 + b^2) - c^2, \quad \mathbb{P}\left(|W| < \frac{1}{2}X^2\right) \geq \frac{3}{4}, \quad \mathcal{A} \equiv \pi$$

References: [1] É. Cartan, Thèse, Paris, 1894. [2] V. I. Arnold, Mathematical Methods of Classical Mechanics, 1989.

PAPER 13 / 論文十三

Riemann-Stieltjes Integration by Parts and Unconditional Mean-Square Dispersion Cancellation

Riemann-Stieltjes 分部積分與 Dirac 預解式二階色散核無條件均方相消大定理

Primary 26A42, 47A55; Secondary 11M06 | MSC 2020

Abstract / 摘要: Using Riemann-Stieltjes integration by parts on $F(t) = \int_0^t |S|^2 = \frac{1}{2} X^2 t$, we prove $\int_0^T t^2 |S|^2 dt = \frac{1}{6} X^2 T^3$. Substituting into $\langle \text{Re } \mathcal{C}_2 \rangle$ yields exact cancellation: $-\frac{1}{48} X^2 T^2 + \frac{1}{48} X^2 T^2 \equiv 0 \cdot X^2 T^2 + \mathcal{O}(XT^2)$, unconditionally.

1. Unconditional Dispersion Cancellation

Theorem 13.1 (Unconditional Mean-Square Dispersion Cancellation)

$$\langle \text{Re } \mathcal{C}_2 \rangle_T = -\frac{1}{8T} \left(\frac{1}{6} X^2 T^3 \right) + \frac{X^2}{16T} \left(\frac{1}{3} T^3 \right) \equiv 0 \cdot X^2 T^2 + \mathcal{O}(XT^2)$$

References: [1] T. J. Stieltjes, Ann. Fac. Sci. Toulouse 8 (1894), 1–122. [2] H. L. Montgomery and R. C. Vaughan, J. London Math. Soc. 8 (1974), 73–82.

PAPER 14 / 論文十四

The Four-Quadrant Epistemic Framework: Resolving Statistics and Bounds in Operator-Zeta Duality

四象限認識論框架：算子-數論對偶體系中無條件統計與條件性界的嚴格劃界大定理

Primary 11M26, 03A05; Secondary 47A10 | MSC 2020

Abstract / 摘要: We establish a 2×2 epistemic classification separating: Quadrant I (Unconditional Mean-Square: $\langle \text{Re } \mathcal{C}_2 \rangle \equiv 0$), Quadrant II (Unconditional Pointwise: $|S| \leq \mathcal{O}_t(e^{X/2 - cX^{1/3}})$), Quadrant III (Conditional Pointwise: $|S| \leq \mathcal{O}_t(X)$), and Quadrant IV (Conditional Consistency).

1. The Four-Quadrant Demarcation

Theorem 14.1 (Conservation of Analytical Difficulty)

$$\log |\det_3(I + V_X R_0(t))| \equiv \frac{1+t^2}{16} X^2 - \frac{t^2}{8} |S(X, t)|^2 + \mathcal{O}_t(X)$$

Crossing from Quadrant II to Quadrant III is logically equivalent to the Riemann Hypothesis.

References: [1] N. M. Korobov, Uspekhi Mat. Nauk 13 (1958), 185–192. [2] I. M. Vinogradov, Izv. Akad. Nauk SSSR 22 (1958), 161–164.

PAPER 15 / 論文十五

De Branges Isometric Chains, Hurwitz Limit Real Zeros, and the Continuum Transference Barrier

de Branges 空間序列等距嵌入、Hurwitz 極限全實零點與連續極限傳遞障壁（Groskin 牆）精確刻畫大定理

Primary 46E22, 30C15; Secondary 47B38, 11M26 | MSC 2020

Abstract / 摘要: De Branges spaces form an isometric chain $\mathcal{H}(E_{X_1}) \hookrightarrow \mathcal{H}(E_{X_2})$. Hurwitz's theorem yields $\text{Zeros}(\mathcal{E}_\infty) = \text{Spec}(\mathcal{D}_\infty) \subset \mathbb{R}$. We formulate the Continuum Transference Barrier (Groskin's Wall): spectral identification $\mathcal{E}_\infty \leftrightarrow \Xi$ is mathematically equivalent to Level III cancellation $|S(X, t)| \leq \mathcal{O}_t(X)$.

1. De Branges Chains and the Continuum Transference Barrier

Theorem 15.1 (Continuum Transference Barrier / Groskin's Wall)

$$\mathcal{E}_\infty(z) \equiv e^{g(z)}\Xi(z) \iff \text{Re} \mathcal{C}_2(X, t) \leq \mathcal{O}_t(X^2) \iff |S(X, t)| \leq \mathcal{O}_t(X)$$

Operator spectral reality is an unconditional theorem; spectral identification with $\zeta(s)$ encapsulates the true arithmetic barrier of the Riemann Hypothesis.

References: [1] L. de Branges, Hilbert Spaces of Entire Functions, 1968. [2] J. B. Conrey and X.-J. Li, Int. Math. Res. Not. 18 (2000), 929–940. [3] D. Groskin, J. Spectr. Theory (2026).